



DriveFace: A Cross-Spectral Through-Glass Face Dataset for On-the-Move Vehicular Border Control

Anjith George¹, Luis S. Luevano¹, Alain Komaty¹, Zeina Al Amine¹, Vidit Vidit¹, Sébastien Marcel^{1,2}

¹Idiap Research Institute, Switzerland ²University of Lausanne (UNIL)
{anjith.george, sebastien.marcel}@idiap.ch



1. Motivation

Growing cross-border mobility motivates **on-the-move authentication**: identifying travellers *directly inside their vehicles*. Face recognition enables passive identification at a distance. However, existing benchmarks do not address practical challenges such as **motion blur, variable illumination, occlusions, window tint, and cross-spectral (RGB→NIR) matching through automotive glass**.

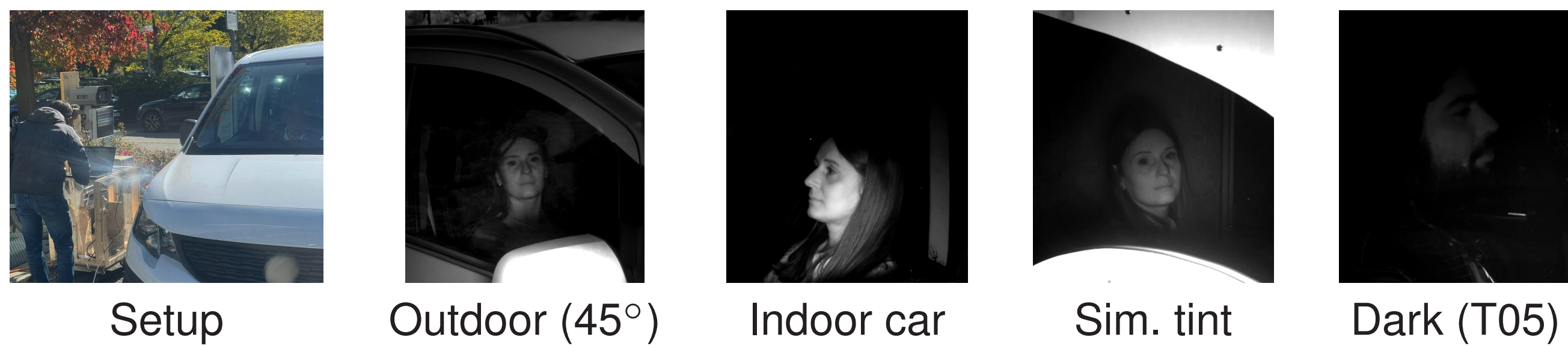
2. Contributions

- **DriveFace dataset**: 70 subjects, two sessions; visible-spectrum smartphone pre-enrollment paired with in-vehicle NIR probes through automotive glass.
- **Cross-spectral, through-glass** RGB–NIR matching at varying tint and viewing angles, stationary and moving.
- **DriveFace-PAD** subset: print, replay, and mask attacks across tint and illumination.
- **Standardized FR + PAD protocols** with rich metadata (tint, illumination, pose, speed) and strong baselines.

3. The DriveFace Dataset

70 subjects **2** sessions **5** tint levels **18–85** age (years)

Workflow: (i) visible-spectrum smartphone pre-enrollment (iPhone 12, Galaxy S9), (ii) in-vehicle NIR captures through glass, (iii) presentation attacks.



Acquisition rig and four NIR probe settings.

4. Demographics

Demographically diverse and gender-balanced (**45.7% female / 54.3% male**), spanning ages 18–85. Apparent skin tone is grouped as light / medium / dark (**51 / 11 / 6** subjects), supporting fairness-aware evaluation.

5. Face Recognition Results

Baselines include AdaFace, LVFace, EdgeFace, and the cross-spectrally adapted **xEdgeFace** across three protocols. The *tinted-glass simulation* is the most challenging setting. **xEdgeFace** outperforms its EdgeFace backbone, with the largest improvement observed in this challenging scenario.

	Model	AUC	EER	VR@FAR=1%	Rank-1
Outdoor	AdaFace	99.45	2.69	96.01	97.42
	LVFace	98.50	5.26	90.17	92.26
	EdgeFace	98.38	5.37	91.35	93.29
	xEdgeFace	98.94	4.18	93.12	94.22
Simulation	AdaFace	96.23	8.26	88.27	90.46
	LVFace	95.19	11.42	79.68	83.95
	EdgeFace	94.09	11.73	82.93	84.72
	xEdgeFace	96.03	8.09	88.63	89.68
Indoor Car	AdaFace	98.59	3.11	96.24	97.11
	LVFace	98.52	5.29	90.23	93.27
	EdgeFace	98.75	4.48	93.92	93.34
	xEdgeFace	97.98	3.47	95.95	96.16

FR performance (%). Best VR@FAR=1% per protocol in **blue**.

6. Effect of Tint Level

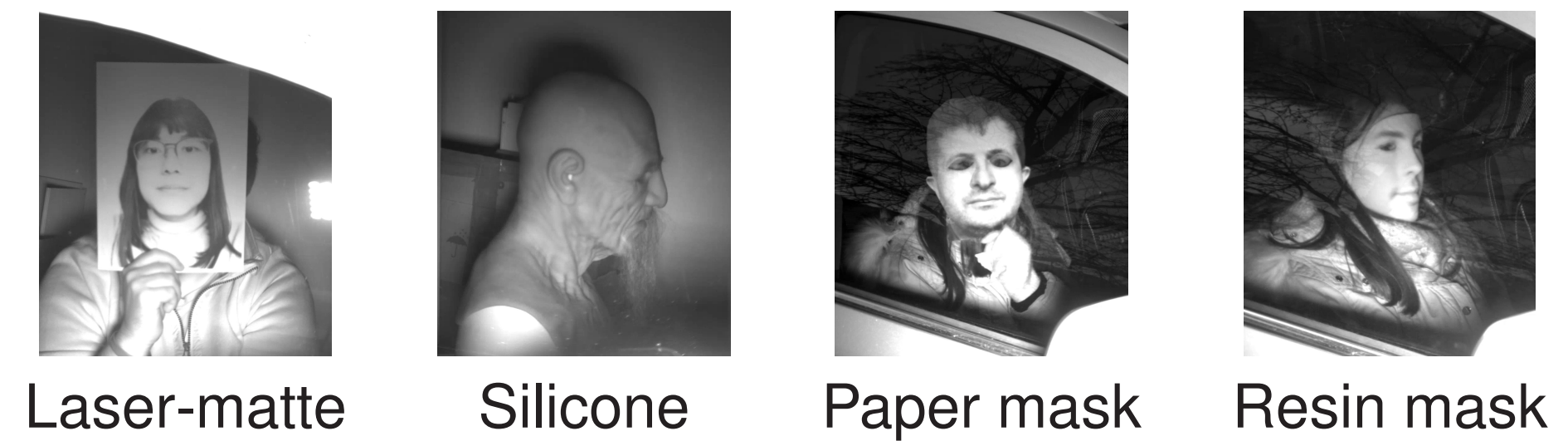
Accuracy generally improves with increasing visible light transmission (VLT), with **clear glass yielding the best performance**, highlighting the impact of window tint.

VLT	AdaFace	LVFace	EdgeFace	xEdgeFace
T05	98.36	98.72	98.42	98.20
T15	98.48	98.09	98.12	98.60
T20	99.31	99.04	98.23	98.72
T35	99.62	99.14	98.75	99.35
clear	100.00	99.86	99.87	100.00

AUC (%) across window tint levels.

7. Presentation Attack Detection

DriveFace-PAD covers print, replay, and mask attacks presented from inside the vehicle and captured through glass in NIR. Known attacks (*grandtest*) are detected reliably, whereas *unseen* attacks reveal a clear **generalization gap**.



	Model	APCER	BPCER	ACER
grandtest	DeepPixBiS	0.00	3.30	1.60
	CLIP (fc)	4.40	2.60	3.50
	DinoV2 (fc)	3.50	1.00	2.20
	EfficientNet-B0	0.10	0.90	0.50
	CLIP (full)	1.60	6.50	4.00
	ConvNeXtV2-T	0.30	1.00	0.70
unseen_mask	DeepPixBiS	91.60	0.00	45.80
	CLIP (fc)	78.80	0.00	39.40
	DinoV2 (fc)	52.30	0.10	26.20
	EfficientNet-B0	54.50	0.00	27.20
	CLIP (full)	83.90	1.60	42.70
	ConvNeXtV2-T	73.10	0.00	36.60
unseen_print	DeepPixBiS	36.70	0.00	18.40
	CLIP (fc)	67.80	0.10	34.00
	DinoV2 (fc)	53.00	0.20	26.60
	EfficientNet-B0	70.70	0.00	35.40
	CLIP (full)	64.70	3.40	34.00
	ConvNeXtV2-T	42.80	0.10	21.50

PAD error rates (%). Best ACER per protocol in **blue**.

8. Key Findings

- Robust FR reaches **96–97% Rank-1** outdoors and indoor-car despite the RGB–NIR gap; the **tinted-glass simulation is hardest** (VR@FAR=1% to ~80%), and accuracy scales with light transmission (clear glass ≈100% AUC).
- **Cross-spectral adaptation (xEdgeFace)** improves over EdgeFace.
- PAD works well for known attacks (**ACER 0.5%**) but challenging on unseen ones (**ACER up to 45.8%**).

9. Resources

Paper page: www.idiap.ch/paper/driveface

10. Acknowledgments

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